Question				Marking details	Marks Available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
4	*(a)*			[Electrical items have] become <u>more</u> [energy] efficient **Don't accept** more efficient energy	1			1		
	(b)			Selection of units used (kWh) = power (kW) × time (h) (1) Substitution: 181 = 130 × time (1) Time = 1 392.3 h (1) [1 392 hours and 18 minutes] Mean number of hours per day = $\frac{1392.3\textbf{ecf}}{365}$ = 3.8 or 4 (1) **N.B.** Answer of 1.39×10^n where n is not equal to 3 award 2 marks Answer of 3.8×10^n where n is not equal to 0 award 3 marks **Alternative 1:** Selection of energy transferred = power × time (1) Substitution: 181 × 3 600 000 = 130 × time (1) Time = 5.01×10^6 [s] (1) [= 1 392.3 h] Mean number of hours per day = $\frac{1392.3\textbf{ecf}}{365}$ = 3.8 or 4 (1) **Alternative 2:** $\frac{181}{365}$ = 0.5 [kWh per day] (1) Selection of units used (kWh) = power (kW) × time (h) (1) Substitution: 0.5 = 130 × time (1) Mean number of hours per day = 3.8 or 4 (1) **Alternative 3:** $\frac{181000}{365}$ = 496 [Wh per day] (1) Selection of units used (kWh) = power (kW) × time (h) (1) Substitution: 496 = 130 × time (1) Mean number of hours per day = 3.8 or 4 (1)	1 1	1 1		4	4	
	(c)			Substitution: 392 = 77.8 × cost per unit (ignore conversions) (1) Cost per unit = 5.0 (1) [p] Answer of 0.05 award 1 mark only	1	1		2	2	
				Question 4 total	**4**	**3**	**0**	**7**	**6**	**0**